

KARNATAKA 1ST PUC BOARD PREPARATION

Chapter 3: Plant Kingdom • Official Blueprint Updated Study Guide

BAIO LEARNING SYSTEM
Narendra Naslai - Founder & CEO
Biology (36) • Max Marks: 70

Updated: 6 Marks Weightage

1. OFFICIAL KARNATAKA 1ST PUC CHAPTERWISE ALLOTMENT

Official Allotment Data (Chapter 3)

- **Chapter Name:** Chapter 3 - Plant Kingdom
- **Teaching Hours Allotted:** 6 Hours (Periods)
- **Official Marks Allotted:** 6 Marks
- **Page Numbers in Official Notes:** Pages 16 – 21
- **Overall Paper Format:** 3 Hours | Max Marks: 70 | Total Allotted Weightage: 105 Marks (including choices)

Expected 6-Mark Question Distribution Configurations

Under the 6-mark allotment, questions can be combined as:

- **Pattern A:** 1 Five-Mark Question (Part D) + 1 One-Mark MCQ/VSA (Part A). **[Most Common]**
- **Pattern B:** 1 Three-Mark Question (Part C) + 1 Three-Mark Question (Part C).
- **Pattern C:** 1 Three-Mark Question (Part C) + 1 Two-Mark Question (Part B) + 1 One-Mark Question (Part A).

2. TOP HIGH-YIELD TOPICS TARGETED FOR 6 MARKS ALLOTMENT

TOPIC 1 [5-Mark / 3-Mark Essential]: Comparative Analysis of Algal Classes

Frequently asked directly as a 5-Mark question in Part D or a 3-Mark question in Part C: "Give the salient differences between Chlorophyceae, Phaeophyceae, and Rhodophyceae."

Feature / Class	Chlorophyceae (Green Algae)	Phaeophyceae (Brown Algae)	Rhodophyceae (Red Algae)
Major Pigments	Chlorophyll a, b	Chlorophyll a, c, Fucoxanthin	Chlorophyll a, d, r-Phycoerythrin
Stored Food	Starch (stored in Pyrenoids)	Laminarin or Mannitol	Floridean Starch
Cell Wall	Inner Cellulose, Outer Pectose	Cellulose coated with Algin	Cellulose, pectin & polysulfate esters
Flagella	2–8, equal, apical	2, unequal, lateral	Absent (Non-motile)
Examples	Volvox, Chara, Spirogyra	Ectocarpus, Laminaria, Sargassum	Polysiphonia, Gracilaria, Gelidium

Topic 2: Heterospory & Seed Habit Precursor (3 or 5 Marks)

Core Question: "What is heterospory? Explain its significance with reference to seed habit in Pteridophytes."

- **Definition (1 Mark):** Production of two distinct types and sizes of spores—smaller **microspores** (male) and larger **megaspores** (female).
- **Examples (1 Mark):** *Selaginella* and *Salvinia*.
- **Evolutionary Significance (1–3 Marks):**
 1. Megaspores germinate into female gametophytes which are **retained on the parent sporophyte**.
 2. The zygote develops into a young embryo **in-situ within the retained female gametophyte**.
 3. This retention and nutritional support of the embryo on the parent plant is a crucial **precursor to the seed habit** in seed plants.

Topic 3: Bryophytes as "Amphibians of Plant Kingdom" & Gametophyte Stages (2 or 3 Marks)

- **Why Amphibians? (1 Mark):** Bryophytes live on damp soil, but ****require liquid water for sexual reproduction**** (flagellated antherozoids swim through water to fertilize the egg in the archegonium).
- **Two Stages of Moss Gametophyte (2 Marks):**
 1. **Protonema Stage:** Creeping, green, branched, filamentous juvenile stage developing directly from a germinating spore.
 2. **Leafy Stage:** Upright, slender axis bearing spirally arranged leaves, developing as a lateral bud from secondary protonema.

Topic 4: Short Notes & Adaptations (1, 2, or 3 Marks)

Economic Importance Notes

- **Algae Hydrocolloids:** Algin (brown algae) and Carrageen (red algae).
- **Agar-Agar:** Obtained from *Gelidium* & *Gracilaria*.
- **Peat Fuel:** Provided by *Sphagnum* moss (high water-holding capacity).
- **Space Food:** Protein-rich *Chlorella* and *Spirulina*.

Gymnosperm Adaptations

- **Leaf Adaptations:** Needle-like foliage, thick waxy cuticle, and **sunken stomata** to minimize transpiration.
- **Root Associations:**
 - **Mycorrhiza:** Fungal association with *Pinus*.
 - **Coralloid Roots:** N₂-fixing cyanobacteria with *Cycas*.

3. MODEL QUESTIONS MATCHED TO 6-MARK BLUEPRINT COMBINATION

Q1. Hydrocolloid 'Algin' is commercially obtained from which class of algae?

1 Mark (Part A MCQ)

Model Answer: (b) Phaeophyceae (Brown algae). [1 Mark]

Q2. Name the nitrogen-fixing cyanobacteria associated with the coraloid roots of *Cycas*.

1 Mark (Part A VSA)

Model Answer: *Anabaena* (or *Nostoc*). [1 Mark]

Q3. What are Gemmae? Mention their function and location in liverworts.

2 Marks (Part B)

Model Answer: Gemmae are green, multicellular, asexual buds produced inside **gemma cups** on the thallus of liverworts (e.g., *Marchantia*). They detach and germinate into new individuals. [2 Marks]

Q4. List three salient anatomical features of Gymnosperms that adapt them to extreme moisture stress. 3 Marks (Part C)

Model Answer:

1. Needle-like leaves in conifers to reduce surface area. [1 Mark]
2. Thick waxy cuticle to reduce cuticular transpiration. [1 Mark]
3. Sunken stomata recessed in pits to trap moisture and restrict water loss. [1 Mark]

Q5. Describe the general characteristics of Division Pteridophyta.

5 Marks (Part D)

Model Answer:

1. **First Vascular Land Plants:** Possess true xylem and phloem tissues. **[1 Mark]**
2. **Dominant Sporophyte:** Plant body is a diploid sporophyte ($2n$), differentiated into true roots, stem, and leaves. **[1 Mark]**
3. **Sporophylls & Strobili:** Sporangia are borne on sporophylls, which may aggregate into compact cones called strobili (e.g., *Selaginella*). **[1 Mark]**
4. **Prothallus Generation:** Spores germinate to form a small, free-living, photosynthetic thalloid gametophyte called a **prothallus**. **[1 Mark]**
5. **Sex Organs & Syngamy:** Bears male antheridia and female archegonia. Water is required for flagellated antherozoids to reach the archegonium. **[1 Mark]**

Blueprint Scoring Tip for Karnataka 1st PUC:

Since Chapter 3 carries **6 Marks** with **6 Teaching Hours**, master the 5-mark distinction table for Algal Classes along with 1-mark objective questions (MCQs/VSAs on examples like *Chlorella*, *Sphagnum*, *Selaginella*).